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SUN SHADE FOR CELL PHONES

Abstract

A portable enclosure for cell phones and electronic devices that provides a sun shade while allowing the transmission of sound there from. The enclosure defines a rectangular shape with a hinge top portion having a deployable sound shade and support panels there within. During use, the hinge top portion is open allowing the placement of a cell phone within the base portion and the sound transmission and support panels hinge from the top portion forming a shade enclosure that contains and direct sound through the sound panel. Selective access is achieved by lifting the hinge sound transmission and shade panel.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Technical Field

[0001] This invention relates to cell phone enclosures that provide shade to prevent the electronic equipment from overheating in the direct sunlight.

2. Description of Prior Art

[0002] Prior art devices of this type have been developed to provide temporary shade for electronic devices by having a shade configuration, see for example U.S. Pat. No. 9,351,415 discloses a folding sun shade having pairs of semi-circular panels that unfold around a cell phone to form an open top partial enclosure for shade.

[0003] U.S. Pat. No. 9,806,756 claims a cell phone case with a shade that retracts into a portion of the phone case and then deploys defining an awning configuration type shade over a portion of the phone.

[0004] U.S. Pat. No. 10,368,622 discloses a glare shield with an open front that fits onto the rear surface of a mobile phone. When unfolded, it overlies the phone screen allowing for access to the power jack's control switches, etc. and thereby reduces the glare on the phone screen in sunlight.

[0005] U.S. Publication 2013/0148204 shows a deployable visor configuration for shielding a flat panel display. It has slidable panels and supports that deploy the panels for selected use.

[0006] U.S. Publication 2014/0349719 discloses an adjustable folding shade for a cell phone with a top and two oppositely disposed side panels forming an open upstanding enclosure to shade the cell phone screen.

SUMMARY OF THE INVENTION

[0007] The shade enclosure for a cell phone and the like has a deployable sound transparent shade panel with end support panels. The enclosure has a phone receiving base with a hinge closure top from which the shade and support panels are hinged.

[0008] The sound transparent shade panel provides a maximum surface for sound transmission therethrough and to shade the phone positioned there within. The end support panels are shaped to fold into the hinge top and then deploy to support the shade panel and top in an angular hinged open position with the sound transparent panel in place for a full shade enclosure that contains and direct sound out through the sound panel.

Description

DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a perspective view of the sound and shade enclosure in open use position.

[0010] FIG. 2 is a perspective view thereof in folded closed transport position.

[0011] FIG. 3 is an end elevational view with the end and sound panel deployment positions shown in solid and broken lines.

[0012] FIG. 4 is an enlarged bottom plan view of the top portion with folded support panels and deployed sound shade panel.

[0013] FIG. 5 is a perspective view of the sound and shade enclosure in open position with unfolded end panels and sound shade panel being open.

[0014] FIG. 6 is a front elevational view of the sound and shade enclosure in deployed open use position.

DETAILED DESCRIPTION OF THE INVENTION

[0015] Referring now to FIGS. 1-5 of the drawings, the sound and shade enclosure **10** of the invention for cell phones and the like can be seen having a rectangular base **11** and a hinged closure top **12**. The based **11** has spaced parallel oppositely disposed upstanding sidewalls **13A** and **13B** with interconnecting opposing end walls **14A** and **14B** and an integral bottom **15** there between. The base **11** defines an open top enclosure for receiving an electronic communication device, cell phone **16**, there within shown in broken lines. The closure top **12** having corresponding dimension to the hereinbefore described base is hinged to the base sidewall **13A**. The closure top **12** has, as

noted, a corresponding upstanding sidewalls **18A** and **18B**, interconnecting end walls **19A** and **19B** and a corresponding interconnected top surface **20** as will be well understood by those skilled in the art.

[0016] As noted, the closure top **12** is hinged to the base **11** by an interlocking hinge configuration **21** along its abutting sidewalls **18A** and **13A**.

[0017] This allows for the closure top **12** to close on the base **11** for phone storage or transport as illustrated in FIG. 2 of the drawings.

[0018] A sound panel **22** is provided of a rectangular dimension with a center sound transparent area **23** within which will allow for the sound transmission there through. The sound panel **22** is pivotally secured to the free edge of the top solid wall **18B** by a continuous hinge **24** that will allow the sound panel **22** to be folded up inside the closure top **12** and unfolded during use as seen in FIGS. 3 and 5 of the drawings.

[0019] A pair of identical end support panels **25** and **26** are hinged at **27** to the inner top surface **28** of the closure top **12** adjacent the respective top end walls **19A** and **19B**, best seen in FIGS. 4 and 5 of the drawings. Each of the end support panels **25** and **26** have a sound panel support edge **25A** and **26A** and a bottom engagement edge **25B** and **26B**. In use, the closure top **12** is opened and the sound panel **22** is first hinged outwardly as seen in FIGS. 3 and 5 of the drawings allowing for the deployment of the respective sound support panels **25** and **26** from inside the closure top **12** interior top surface **27** indicated by directional arrows A in FIG. 5 of the drawings.

[0020] The closure top **12** is then partially closed with the hereinbefore disclosed engagement of the end support panels **25** and **26** in the base **11** for support and the sound panel **22** is then hinged down to engage on the now angularly positioned sound panel support edges **25A** and **26A** defining a shade enclosure for the cell phone **16** there within as seen best in FIG. 1 of the drawings.

[0021] It will be evident from the disclosure that the sound panel **22** will allow the now shaded cell phone **16** to be heard clearly outside the sound and shade enclosure **10**. It will also be apparent that it will allow for ease of phone access by simply lifting up the freely hinged sound panel **22** for quick access and removal of the phone as needed.

[0022] It will thus be seen that a new and novel sun shade and sound enclosure for cell phones and electronic devices have been illustrated and described and it will be apparent to those skilled in the art that various changes and modifications may be made thereto without departing from the spirit of the invention.

Claims

1. A shade and sound enclosure for a cell phone comprising, a base enclosure portion and a top enclosure portion hinged together, said base enclosure portion having upstanding sidewalls and interconnecting end walls, said top enclosure portion having upstanding sidewalls at interconnecting end walls there between, a sound panel hinged from one of said top portion's sidewalls movable from a first position to a second position, a pair of oppositely disposed sound panel supports hinged from said top portion.
2. The shade and sound enclosure set forth in claim 1 wherein said base enclosure and said top enclosure are of equal registration dimension and aligned with one another when in hinged closed position.
3. The shade and sound enclosure set forth in claim 1 wherein said sound panel and said sound panel supports are configured so that they can fold within said top enclosure portion.
4. The shade and sound enclosure set forth in claim 1 wherein said sound panel and said sound panel supports upon deployment will define a shade and sound transparent enclosure.
5. The shade and sound enclosure set forth in claim 1 wherein the dimensions of said sound panel is less than the known dimension of said top enclosure portion.
6. The shade and sound enclosure set forth in claim 1 wherein said sound panel has a sound

transparent surface.

7. The shade and sound enclosure set forth in claim 1 wherein said base and top enclosure portions are of a rectangular dimensional configuration.

8. The shade and sound enclosure set forth in claim 1 wherein said sound panel supports folding sequentially in both collapsing and deployment positions.

9. The shade and sound enclosure set forth in claim 1 wherein said top enclosure portion is movable from a first closed position in registration on said base enclosure portion to a second open position in angular orientation to said base enclosure portion.

10. The shade and sound enclosure set forth in claim 1 wherein said sound panel first position is folded within said top enclosure portion and said second position is hinged from said top enclosure to one of said base portion sidewalls and said respective sound panel supports.
